Frazil deposition under growing sea ice

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[1] Platelet ice may be an important component of Antarctic landfast sea ice. Typically, it is found at depth in first-year landfast sea ice cover, near ice shelves. To explain why platelet ice is not commonly observed at shallower depths, we consider a new mechanism. Our hypothesis is that platelet ice eventually appears due to the sudden deposition of frazil ice against the fast ice-ocean interface, providing randomly oriented nucleation sites for crystal growth. Brine rejected in plumes from landfast ice generates stirring sufficient to prevent frazil ice from attaching to the interface, forcing some of it to remain in suspension until ice growth rate and brine rejection slow to the point that frazil can stick. We calculate a brine plume velocity and match this to frazil rise velocity. We consider both laminar and turbulent environments. We find that brine plume velocities are generally powerful enough to prevent a significant range of frazil sizes from sticking in the case of laminar flow and that, in the turbulent case, there may be a critical ice thickness at which most remaining circulating frazil suddenly settles.

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1. Introduction

[2] During the early freezing of the ocean, small discs of ice (frazil) accumulate near the surface and eventually stick together to form a relatively solid layer of small randomly oriented crystals, usually about 10 cm thick. Then further growth favours crystals with their fastest growth plane near vertical. Columnar ice results, with large crystals and nearly horizontal c axes [e.g., *Eicken*, 2003; *Weeks and Ackley*, 1986].

[3] Another type of ice is commonly observed in Antarctic landfast sea ice, especially near ice shelves. This is platelet ice, consisting of various-sized, flat, plate-like crystals with disordered c axis orientations. Platelets have been defined as dendritic single crystals with no inclusions [*Moretskii*, 1965; *Jeffries et al.*, 1993]. They grow up to 200 mm in diameter, and are up to about 3 mm thick. They have also been called frazil ice, but there is a variety of terminology (see *Smith* [2001], for a full discussion). They are found both freely circulating as individual crystals in supercooled seawater beneath a solid sea ice cover, and incorporated into the structure of landfast sea ice. In this paper, we use the term frazil to refer to small ice crystals that drift in the water columns and display as crystals up to about 5 mm in radius when found in a sea ice core, without any implications for genesis. We use the term platelet to refer to crystals that originate in the water column and have a genesis associated with interaction with an ice shelf. Platelets may become attached to and possibly incorporated into the sea ice cover, and there is no implication that platelets are small. There is also no suggestion that frazil cannot grow into platelets, in a supercooled water column or while attached to the sea ice cover.

[4] In the McMurdo Sound region of the Ross Sea, platelet ice is typically observed in fast ice cores at depths greater than about 1 m [*Jones and Hill*, 2001; *Gow et al.*, 1998; *Smith et al.*, 1999, 2001; *Jeffries et al.*, 1993; *Leonard et al.*, 2006]. In such regions, turbulent tidal ocean flows, episodically supercooled from early winter onward and carrying billows of frazil ice, complicate the picture [*Leonard et al.*, 2006; *Jenkins and Bombosch*, 1995; *Purdie et al.*, 2006]. Platelet ice incorporated into fast ice correlates strongly with the presence of frazil in the underlying water column [*Leonard et al.*, 2006].

[5] A number of hypotheses have been proposed to explain why platelet ice is commonly found in sea ice cores near ice shelves in Antarctica at ice depths greater than about 1 m, but is largely absent at shallower depths. The presence of ice shelves nearby is known to be important [*Crocker and Wadhams*, 1989; *Holland and Jenkins*, 1999; *Jenkins and Bombosch*, 1995; *MacAyeal*, 1984], with supercooled plumes leaving the ice shelf at neutral buoyancy levels and moving out under nearby sea ice. Frazil ice in these plumes grows, collides and provides secondary nucleation sites, and rises out of suspension due to buoyancy [*Daly*, 1984]. Rise velocities depend on the volume of the disc-shaped frazil crystals [*Gosink and Osterkamp*, 1983], as well as disc orientation and turbulence (this work). The platelet ice found in landfast sea ice may be the result of

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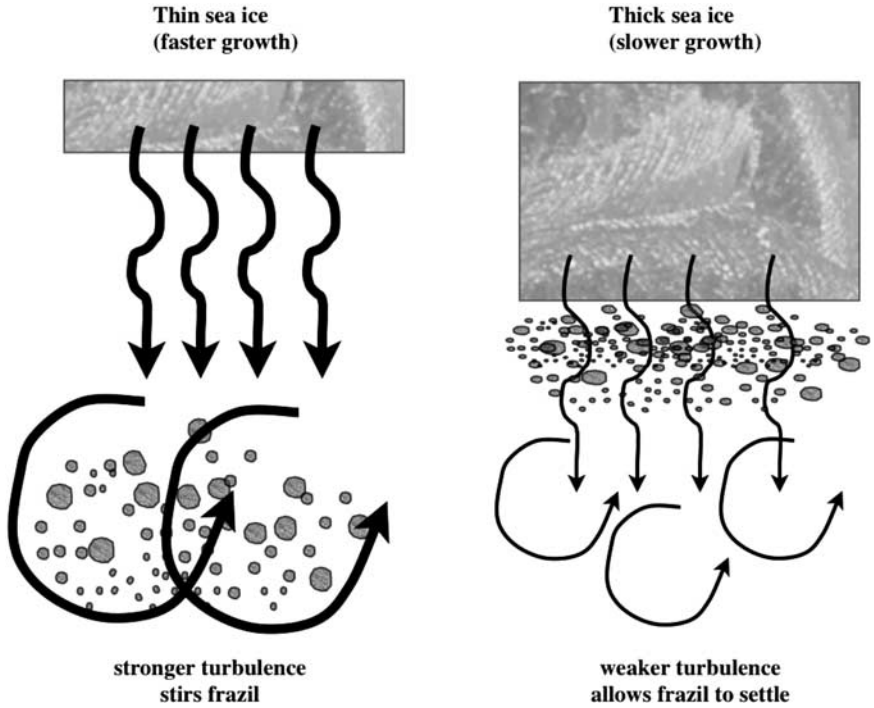

Figure 1. A conceptual sketch illustrating stronger brine-driven turbulence under thinner sea ice, mobilizing frazil ice in the ocean.

frazil that has grown entirely in the supercooled plume before rising out of suspension to attach to the fast ice; or it may have grown entirely at the fast ice interface; or a combination of these two mechanisms may occur [*Veazey*, 1994; *Veazey et al.*, 1994; *Gow et al.*, 1998; *Smith et al.*, 1999, 2001]. The random orientation of platelets observed in landfast ice suggests they grow mostly in place [*Gow et al.*, 1998], rather than arriving as full-sized crystals, since a large disc would be expected to settle against the fast ice interface with its c axis almost vertical. However, this argument assumes that the ice-ocean interface is flat compared with the disc size, and that a disc arrives at the interface alone. The delayed appearance of platelets in fast ice in some areas may be due to the residence time of frazil ice in the supercooled plume; or it may be due to the time required to remove sensible heat from the ocean immediately below the fast ice [*Crocker and Wadhams*, 1989]; or it may be due to the new hypothesis to be outlined here.

[6] The transition that is sometimes seen in Antarctic sea ice, from columnar to platelet ice, has a compelling parallel in the columnar-to-equiaxed crystal transition seen in binary alloy solidification [*Badillo and Beckermann*, 2006; *Ghosh*, 2001]. Often, with cooling from the outside of a casting of a binary alloy, columnar crystals grow horizontally from the walls toward the center of the casting as it sets, and small crystals grow in the convecting melt where it has not yet solidified. In the solid casting, a transition from columnar to equiaxed crystals (randomly oriented like our platelets) is seen, associated with the location of the convection front when it sets. Convection of the melt containing growing crystals occurs in the initial stages, and randomizes the crystal orientations. Eventually the viscosity of the convecting melt increases and convection ceases, freezing the randomized crystals in place.

[7] There is a key difference between these two applications of solidification of a binary melt; in alloy solidification, the process continues until all of the melt has frozen, and the transition to platelet or equiaxed crystals is associated with the mushy zone taking over the melt region. In the case of sea ice, there is plenty of melt remaining below fast ice at the end of winter; on the other hand, buoyancy effects act to bring frazil crystals up against the fast ice, while turbulence acts to keep them in suspension [*Svensson and Omstedt*, 1998]. It is clear that in the case of alloy solidification, the equiaxed crystals must eventually become part of the solid. It is not so clear in the case of sea ice if and when this transition occurs, and why it should be sudden.

[8] In this paper we consider the role played by brine rejection from growing fast ice, and how it might interact with frazil in the water column. Narrow plumes of cold brine flow down from growing fast ice into the ocean, and slower warmer wider regions of seawater flow up into the permeable lower levels of the fast ice in return. These brine plumes, if they flow rapidly enough, may keep frazil in suspension, so that fast ice is composed of congelation ice. We investigate the possibility that, as the sea ice thickens, the flow in brine plumes may weaken sufficiently that frazil can settle against the fast ice, and become incorporated into it. The situation is summarized in the conceptual sketch in Figure 1.

[9] The spacing between brine plumes will be less than or equal to the spacing between the small brine tubes seen in cores taken from sea ice [*Lake and Lewis*, 1970], since each such tube will eject brine in a plume. There may be additional plumes originating, for example, at the tips of dendrites. The small brine tubes have diameters of about 0.4 mm, are spaced about 1.5 mm apart, and occupy about 5% of the surface area of the sea ice.

Table 1. Symbols and Constants

Symbol	Value	Units	Definition
B		$m^3\ s^{-3}$	buoyancy flux
C_d	1.3×10^{-3}		drag coefficient
F	1 or 3/2		disc orientation factor
g	9.8	$m\ s^{-2}$	gravitational acceleration
h		m	sea ice thickness
r	0.004–5	mm	frazil disc radius
Re	$\frac{UL}{\nu}$		Reynolds number
S_i	5	psu	salinity of sea ice
S_w	35	psu	salinity in mixed layer
T_a	−25	°C	air temperature
T_f	−2	°C	freezing temperature
u_*	0.001–0.030	$m\ s^{-1}$	friction velocity
V		$m\ s^{-1}$	rise velocity
W_*		$m\ s^{-1}$	convective velocity scale
z_i	25–75	m	depth of mixed layer
ϵ	0.01		aspect ratio $t/(2r)$
κ	0.4		von Karman's constant
μ	1.88×10^{-3}	Pa s	dynamic viscosity
ν	1.8×10^{-6}	$m^2\ s^{-1}$	kinematic viscosity
ϱ_i	925	$kg\ m^{-3}$	density of sea ice
ϱ_w	1028	$kg\ m^{-3}$	density of seawater
ϱ_{ws}	0.81	$kg\ m^{-3}\ psu^{-1}$	$\frac{d\varrho_w}{dS}$

[10] Frazil and platelet disc sizes are variously reported in the literature. *Holland and Feltham* [2005] describe frazil discs as having radii in the range 0.01–10 mm. *Martin* [1981] in his review of field observations of river frazil notes a maximum radius of 0.1–2.5 mm. Laboratory observations indicate radii up to 10 mm [*Smedsrud*, 2001], and *Dieckmann et al.* [1986] trawled platelets with an average radius of 10 mm in a net with a mesh size of 10 mm. Modeling work by *Bombosch* [1998] also supports disc radii up to 10 mm.

[11] *Daly*'s [1984] theory has a distribution of frazil sizes that falls off very rapidly for radii greater than 1 mm. Other modelers use radii ranging from 0.003 to 5 mm [*Jenkins and Bombosch*, 1995; *Martin*, 1981; *Svensson and Omstedt*, 1994]. Aspect ratio (disc thickness over diameter) is typically $\epsilon \approx 0.01$, but may be as large as 0.1. The size spectrum of Figure 9 in *Smedsrud and Jenkins* [2004] arises out of their modeling, and has a peak between 1 and 2 mm in size, with a range up to 3 mm. The largest disc size in the modeling described by *Svensson and Omstedt* [1998] is 10 mm, and while the numbers of such discs may be as high as 10^5 per m^3 at 1 m depth, this is a factor of 10^6 smaller than the peak disc numbers at about 0.01 mm radius.

[12] These wide ranges and variations in size distributions do not directly affect our calculations, as we make no specific assumptions about frazil sizes under sea ice in these calculations. We do assume for the purposes of our conclusions, that relatively few discs are larger than 5–10 mm in radius, and that most discs are smaller than 3 mm in radius. There is strong support in the above literature for these assumptions.

[13] If the frazil is of a similar length scale to the roughness of the mushy zone/seawater interface, or if the amount of frazil (previously well stirred by turbulence) that can settle against the interface is thick enough, then an episodic reseeding of the fast ice interface with frazil ice with random c axis orientations and crystal sizes may be associated with times of low turbulence, due to low tidal shear flow. Tidal records from McMurdo Sound, Antarctica [*Goring and Pyne*, 2003] suggest a 14-day period, associated with the spring/neap tidal cycle. Times of low shear are anticipated due to both this cycle, and to the diurnal tidal cycle.

[14] Subsequent fast-ice growth during more vigorous shear flow will be from these newly adhered randomly oriented seed crystals, and might be expected to have the characteristics of platelet ice, with sizes limited by the time until the next low shear episode allows new seeds to settle, and also limited by the geometric exclusion of crystals with more vertically oriented c axes that eventually (in the absence of reseeding) leads to columnar ice with horizontal c axes.

[15] In the following sections, we use solid sea ice growth rates measured in the field, to calculate brine plume velocities as a function of sea ice thickness. Frazil rise velocities in still and turbulent water are used to calculate the plume velocity that will mobilize frazil ice of a given radius, leading to a relationship between fast ice thickness and the maximum size of frazil that is kept in suspension. In the case of a turbulent ocean, we find there is a sudden change in the size of frazil that can be kept in suspension, at a critical fast ice thickness. Symbols and constants are summarised in Table 1.

2. Brine Plume Velocities

[16] We seek a representative velocity for brine plumes, to compare with the rise velocity of frazil crystals. The velocity of choice [*McPhee and Morison*, 2001] is the Deardorff [*Deardorff*, 1970] or free convection velocity scale W_*, which is a measure of the velocity expected to develop in an individual turbulent plume driven by a localized buoyancy source.

[17] The Deardorff velocity may be written [e.g., *McPhee and Morison*, 2001]

$$W_*^3 = B\lambda, \tag{1}$$

where λ is a measure of the size of the dominant eddies, inferred from measurements [*McPhee and Morison*, 2001] to approach κz_i in the limit of negligible shear, where κ = 0.4 is von Kármán's constant. The term z_i is the thickness of the oceanic layer that is well mixed by the rejected brine, that is, the depth to the pyncnocline, typically about 50 m in McMurdo Sound in midwinter [*Leonard et al.*, 2006], but reaching 75–100 m in the Weddell Sea. The simulations of *Mellor et al.* [1986] also use a depth of 50 m for the well-mixed region. In general, this depth will slowly grow, depending on the salinity versus depth curve that was in place in the seawater before the sea ice began to form, and the pyncnocline at the bottom will be gradually eroded to greater depths. Since the mixed layer depth is a difficult concept in McMurdo Sound because of the weak stratification in the water column, and since mixing into the pyncnocline is a poorly understood problem [*McPhee and Morison*, 2001], we assume a constant z_i = 50 m here, but we later explore the sensitivity of our results to variations in z_i by also considering values of 25 and 75 m.

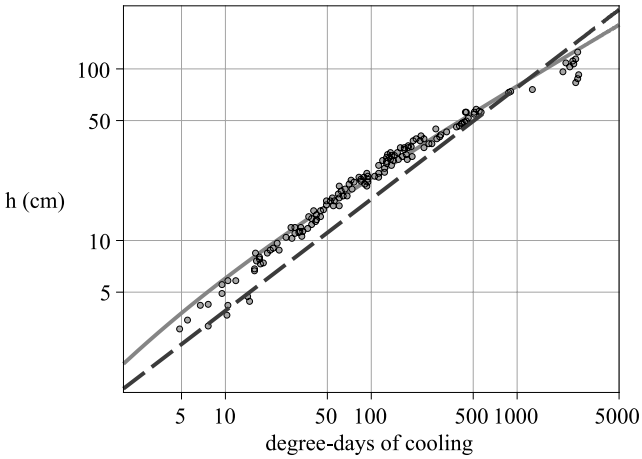

Figure 2. Data and a curve fitted by eye, for the thickness of snow-free ice versus the number of degree-days of cooling, taken near Greenland, after *Anderson* [1961]. Anderson's data is shown as filled circles and his fit as a solid curve through them, on a log-log plot. Also shown is a dashed line, representing data from the shipping lane in McMurdo Sound from the work of *Purdie et al.* [2006].

[18] B is the buoyancy flux [*Turner*, 1973], here taken to be positive downward,

$$B \equiv \frac{g\langle \varrho'_w w' \rangle}{\langle \varrho_w \rangle}$$

where $\langle \varrho'_w w' \rangle$ is the average downward flux of random variations ϱ'_w in fluid density at the ice/water interface, and $\langle \varrho_w \rangle$ is the average density of seawater in the mixed layer.

[19] A good fit to the dependence of the density of seawater on salinity S_w (psu) at the freezing point [*Unesco*, 1981] is

$$\varrho_w \approx 999.84 + \left(\frac{d\varrho_w}{dS}\right) S_w, \quad \text{where} \quad \frac{d\varrho_w}{dS} \equiv \varrho_{ws} \approx 0.81,$$

so that

$$B \approx \frac{\varrho_{ws} g \langle w' S'_w \rangle}{\langle \varrho_w \rangle}. \tag{2}$$

[20] The vertical flux of salt due to the freezing of the fast ice cover may be written in terms of its rate of growth $\dot{h}$ in the absence of frazil accretion, to get

$$\langle w' S'_w \rangle = \dot{h} \frac{\varrho_i}{\varrho_w} (S_w - S_i), \tag{3}$$

where $\varrho_i \approx 925$ kg m^{-3} is the density of the sea ice, and S_i is the salinity of the sea ice, typically proportional to the ocean salinity [*Schmidt et al.*, 2004], so that $S_i \approx 0.14\ S_w$. We approximate the density ϱ_w of brine immediately below growing sea ice by the density of the well-mixed layer (1028 kg/m^3), and the salinity S_w by the salinity of the well-mixed layer, typically about 35 psu in McMurdo Sound.

[21] Hence we have the convective velocity scale for brine plumes under growing sea ice W_*, given in terms of ice growth rate $\dot{h}$ by

$$W_*^3 = 0.344 g \varrho_{ws} \frac{\varrho_i}{\varrho_w^2} S_w z_i \dot{h}. \tag{4}$$

[22] The associated Reynolds number is (with z_i = 50 m, and at an ice thickness of 1 m, when $\dot{h} \approx 0.1$ μm/s, so that $W_* \approx 5$ mm/s)

$$\mathrm{Re} = \frac{UL}{\nu} \approx \frac{5 \times 10^{-3} \times 50}{1.8 \times 10^{-6}} \approx 1.4 \times 10^5.$$

This is consistent with turbulent plume flow, as it is much bigger than the critical value of about 200 for buoyant plumes [*Snyder*, 1981].

[23] *Niedrauer and Martin* [1979] measure a brine flow rate of 0.25 mm/s, in an experiment with $z_i \approx 0.1$ m and $\dot{h} \approx$ 1–5 μm/s. Then equation (4) gives $W_* \approx 2$ mm/s, which is a factor of eight higher than their observed brine flow rate. This is likely to be due to the effect of the plastic sidewalls in their experiment, which are sufficiently close together to damp plume flows. With a spacing of just 1.6 mm between walls, the apparatus is essentially a Hele-Shaw cell, and plume flow is not turbulent. Indeed, a Reynolds number calculation gives

$$\mathrm{Re} = \frac{UL}{\nu} \approx \frac{0.25 \times 10^{-3} \times 1.6 \times 10^{-3}}{1.8 \times 10^{-6}} \approx 0.25,$$

which is much less than 200, so that plume flow is dominated by viscous effects at this Reynolds number.

[24] *Voropayev and Fernando* [1999] grow sea ice in a box measuring 30 × 61 cm^2 in cross section and 60 cm in depth. They have a depth to pyncnocline of 0.2 m, and ice growth rates of 1–4 μm/s. Then our velocity scale predicts brine plume velocities $W_* \approx$ 2–3 mm/s in their experiments, consistent with their calculations of brine plume velocities of 1–4 mm/s. The Reynolds number for their experiment is Re $\approx$ 400, just high enough to be turbulent.

3. Ice Growth Rate

[25] We need ice growth rates to calculate brine rejection rates. Measurements of the thickness of snow-free ice [*Anderson*, 1961] in Thule, Greenland, provide direct estimates for growth rate as a function of ice thickness and air temperature. Anderson fitted the formula

$$h^2 + 0.051h = 0.775 \times 10^{-8}\theta \tag{5}$$

where h is in m,

$$\theta \equiv \int_0^t (T_f - T_a)\,dt$$

is the number of degree-seconds of cooling, T_f is the temperature of the interface between ice and ocean (at freezing point), and T_a is the air temperature, both in °C. This formula, which is derived from a simple model of sea

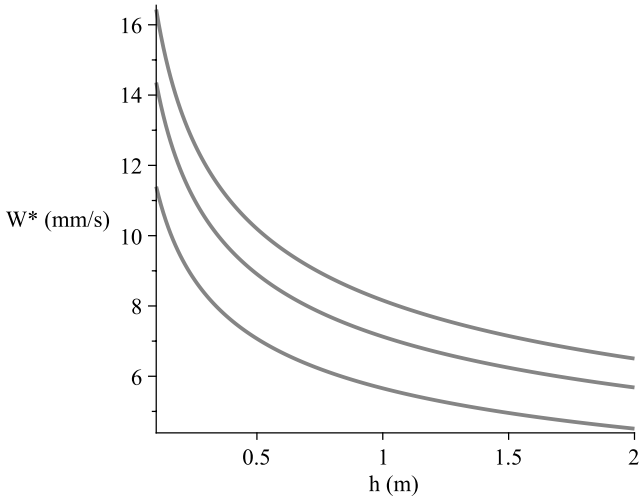

Figure 3. Calculated values of brine plume velocity versus sea ice thickness. There are three curves, the lower one is for z_i = 25 m, the middle one for z_i = 50 m, and the upper one for z_i = 75 m.

ice growth [see also *Lepparanta*, 1993; *Voropayev and Fernando*, 1999], is illustrated together with Anderson's data in Figure 2. It follows from Anderson's formula that

$$\dot{h} = \frac{0.775 \times 10^{-8}}{2h + 0.051}\left(T_f - T_a\right). \tag{6}$$

[26] Measurements in McMurdo Sound [*Leonard et al.*, 2006] give an eventually more rapid growth, as also illustrated in Figure 2. This may be due to delayed accretion of frazil ice to the fast ice-ocean interface, leading to the incorporated platelet ice seen in cores. We use Anderson's data as a proxy for ice growth rate in the absence of frazil accretion, since frazil accretion does not in itself lead to brine rejection from the fast ice interface, and since incorporated platelets are not observed in fast ice cores from the Arctic.

[27] Air temperatures of −25°C and a freezing temperature of −2°C will be used in Anderson's empirical formula, when illustrating typical growth rates. This leads to the values of W_* graphed against sea ice thickness in Figure 3, for mixed layer depths of 25, 50 and 75 m.

4. Frazil Mobilization Criteria

[28] Two mechanisms for mobilizing frazil are considered, a Shields criterion from sedimentation theory for turbulence to mobilize frazil, and brine plume velocity matching frazil rise velocity. The second mechanism varies with ice thickness, so has the possibility of explaining delayed emplacement of frazil in the ice cover, while the first does not, unless the only source of turbulence is the brine plumes. Both mechanisms interact with each other, since brine plumes contribute to the level of turbulence, and turbulence is known to modify the rise velocity of frazil.

4.1. Shields Criterion

[29] *Jenkins and Bombosch* [1995] use a Shields criterion for deposition of frazil rising in a turbulent plume under an ice shelf. They note that their formulation is only appropriate for a grain Reynolds number greater than ~1, corresponding to a frazil disc radius greater than ~1 mm. Frazil sizes can be much smaller than this. Extensions to smaller grain Reynolds numbers have been made [*Miller et al.*, 1977]. *Cao et al.* [2006] report an explicit relationship covering a wide range of particle sizes, which we adopt. This relationship says that all frazil with radii less than or equal to r are mobilized when the friction velocity (u_*, a measure of the turbulent velocity scale) exceeds the critical value

$$u_*^c = \sqrt{2rg(1 - \varrho_i/\varrho_w)\theta_c},$$

where the critical value of the Shields parameter has been carefully matched to experimental data to get

$$\theta_c = \begin{cases} 0.1414\mathrm{Re}_g^{-0.2306}, & \mathrm{Re}_g < 6.61 \\ \dfrac{\left[1 + \left(0.0223\mathrm{Re}_g\right)^{2.8358}\right]^{0.3542}}{3.0946\mathrm{Re}_g^{0.6769}} & , 6.61 \le \mathrm{Re}_g \le 282.84\,, \\ 0.045, & \mathrm{Re}_g > 282.84 \end{cases}$$

and $\mathrm{Re}_g = 2r\sqrt{2rg(1 - \varrho_i/\varrho_w)}/\nu$ is the grain Reynolds number.

[30] A graph of u_*^c versus r is shown in Figure 4, with u_*^c taking values up to nearly 20 mm/s for the largest frazil sizes we expect. This may be compared with the range of values u_* = 1–30 mm s^{-1} measured by *McPhee et al.* [1999] in 1 m thick winter pack ice in the Weddell Sea. Typical values are 10–20 mm s^{-1}. At the low end of this range, some larger frazil discs can settle against the solid ice cover. At values above 15 mm s^{-1}, very little frazil in the size range of interest can settle out.

[31] Brine plumes also contribute to friction velocity. However, as we saw in Figure 3, brine plume velocities are of the order of 5–15 mm s^{-1}, giving an additive

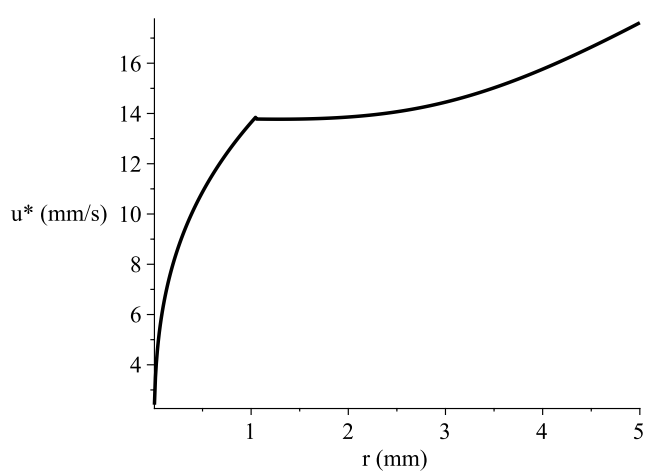

Figure 4. The Shields criterion for critical value of friction velocity (mm/s) against frazil disc radius (mm). When friction velocity takes the critical value, all frazil discs up to the indicated radius are mobilized by the turbulence.

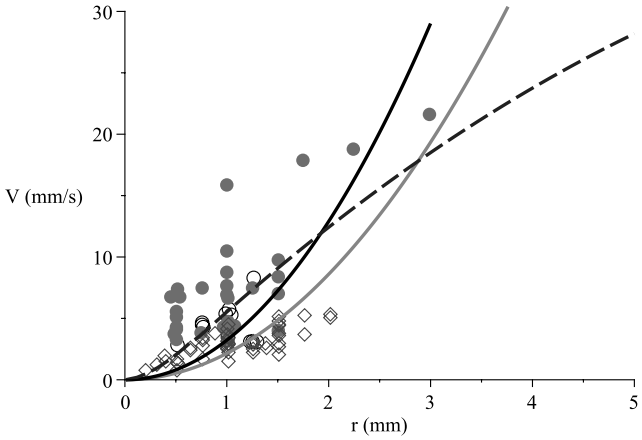

Figure 5. Frazil rise velocities V versus disc radius r, in still water. Filled discs are field observations by *Gosink and Osterkamp* [1983], circles are their laboratory data, and diamonds are laboratory data from *Wuebben* [1984]. The dashed line is the Morse and Richard data fit given by equation (7). The solid lines are from a Stokes flow analysis, for a disc rising face up (the lower curve) and edge up (the upper curve) in still water.

contribution of $\sqrt{C_d}W_* \approx 0.2$–$0.5$ mm/s to the friction velocity [e.g., *Beljaars*, 1995; *Grachev et al.*, 1998], where the drag coefficient $C_d \approx 1.3 \times 10^{-3}$ [e.g., *Holland and Jenkins*, 1999]. So brine plumes may augment a shear-induced turbulent environment by only 1–2%, and in the absence of shear are too weak to mobilize any but the very smallest frazil through the Shields criterion, so we do not consider this further.

[32] Brine plumes differ crucially from shear flow, in that they produce a net downflow, which rising frazil must conquer in order to settle out. So we now consider the rise velocity of frazil, in an environment where turbulence may take various values, from zero to those mentioned above.

4.2. Frazil Rise Velocity

[33] Measurements of observed frazil rise velocities in fresh water [*Gosink and Osterkamp*, 1983] have been modeled by matching buoyancy and drag forces, where the drag force is given by a quadratic dependence on rise velocity, modified by a velocity-dependent coefficient of drag. Such a dependence allows the theory to cover both linear and nonlinear drag regimes, since the drag coefficient becomes inversely proportional to velocity in the low Reynolds number limit (Stokes flow limit), making the overall dependence linear in rise velocity. The frazil disc sizes they consider have radii in the range 0.5–3 mm, with Reynolds numbers from 1 to 75, and they note that Stokes' theory is not appropriate for the larger sizes (Re > 5). However, their treatment of frazil disc orientation is very simple; they replace the disc with a sphere of the same radius, for the purposes of drag calculations.

[34] The rise velocities of frazil observed by *Gosink and Osterkamp* [1983] and shown here in Figure 5 lie in the range 5–15 mm s^{-1}, which matches the range of brine plume velocities calculated in section 2. This already indicates, regardless of the model chosen for frazil rise velocity, that brine rejection is of the right order of magnitude to mobilize frazil rising in a nonturbulent (laminar) environment.

[35] The very recent work of *Morse and Richard* [2009] is prompted by their study of acoustic measurements of frazil concentrations in a flowing river. They conduct an extensive and detailed review of frazil rise velocities, and they also consider work done on the terminal velocities of droplets and crystals falling through the atmosphere. Their preferred model is a power law fit by *Khvorostyanov and Curry* [2002] with continuously varying parameters over the size spectrum found in the atmosphere, which spans both Stokes and turbulent flow regimes, and is valid for Reynolds numbers up to 5000. *Morse and Richard* [2009] use data from the field and lab work of *Gosink and Osterkamp* [1983], and lab data from *Wuebben* [1984], to evaluate frazil rise models. They give a simple approximation (also valid for Re up to 5,000) to their preferred model for frazil discs which can be written as

$$W_r = \begin{cases} 6.229r^{1.621}, & r \leq 0.635 \text{ mm} \\ -0.412r^2 + 8.138r - 2.024, & r > 0.635 \text{ mm} \end{cases} \tag{7}$$

where W_r is the rise velocity in mm s^{-1} and r is disc radius in mm.

[36] The rise velocity given by this approximation is graphed together with the data of *Gosink and Osterkamp* [1983] and *Wuebben* [1984], in Figure 5. It is a good fit to the data. However, the scatter of data allows a number of models.

[37] For smaller frazil discs, with radii in the range 0.004–1 mm, it is appropriate to calculate the frazil rise velocity V using a Stokes flow analysis, since the Reynolds number Re = $2Vr/\nu$ is less than 1. An advantage of the Stokes flow analysis is that it allows us to compute explicitly the effect of disc orientation and aspect ratio, on rise velocity.

4.3. In Still Water

[38] When discs move under the influence of their own buoyancy in a still viscous fluid, they exhibit a behavior that ranges from steady movement with the surface normal aligned with gravity, through periodic motion to chaotic or tumbling motion [*Field et al.*, 1997], depending on their size and relative density. For ice discs with radii up to 10 mm and an aspect ratio of 0.01 rising in still water, the steady behavior is observed, with the largest surface area presenting at right angles to the flow (that is, with the c axis vertical, as also observed in frazil experiments in salt water by *Ushio and Wakatsuchi* [1993]). Stokes flow at velocity V past a thin disc of radius r in this orientation [*Lamb*, 1932] leads to the viscous drag force

$$6\pi\mu RV \tag{8}$$

where $R = 8r/(3\pi)$, and $\mu \approx 1.88 \times 10^{-3}$ Pa s is the dynamic viscosity of seawater. Matching this force with the buoyancy force $g(\varrho_w - \varrho_i)2\epsilon\pi r^3$ gives

$$V = \frac{\pi g(\varrho_w - \varrho_i)\epsilon r^2}{8\mu} \approx 2.1 \times 10^3 r^2. \tag{9}$$

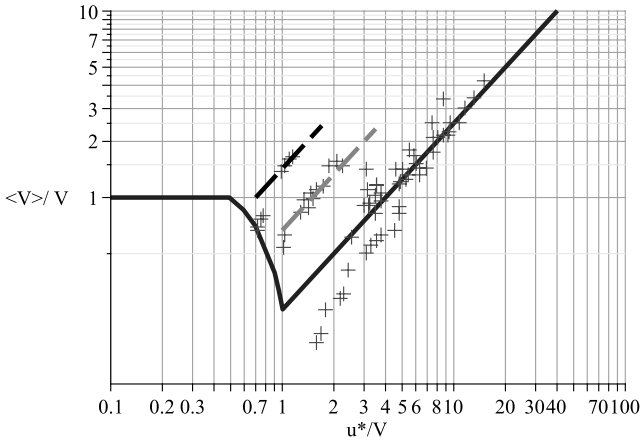

Figure 6. Observed mean diesel droplet rise velocities $\langle V \rangle$ versus friction velocity u_*, both normalized on the rise velocity V in still water, data from *Friedman and Katz* [2002] as crosses, and our hand-fitted behaviors shown as solid and dashed lines.

[39] In the turbulent environment induced in the ocean by shear flow and plumes, it is to be anticipated that the above analysis for discs rising in still water should be modified. Two consequences are considered here, that the orientation of the disc is not necessarily with the c axis vertical, and that turbulent vortices directly affect the rise velocity.

4.4. Disc Orientation in Turbulent Water

[40] When a disc rises with the c axis horizontal (that is, thin edge up), there is a different value $R = 16r/(9\pi)$ in the drag force given by expression (8) [*Lamb*, 1932]. Since this is 3/2 times the R value for discs rising face-up, matching the drag force with buoyancy gives a rise velocity that is exactly half again as large as when discs rise face-up (equation (9)). So we will allow for the possibility that rise velocity is up to 50% larger, in our analysis, since discs in a turbulent environment may be at some time-varying angle between face-up and edge-up, and the actual velocity is expected to vary between these two values. We can write the rise velocity for both orientations in the form

$$V = \frac{\pi g(\varrho_w - \varrho_i)F\epsilon r^2}{8\mu}, \tag{10}$$

where the disc orientation factor $F = 1$ for discs rising face-up, and $F = 3/2$ for discs rising edge-up (obtained directly from the ratio of the two different R values).

[41] The effect of a 50% increased aspect ratio ϵ can also be seen from the effect of changing F, directly from Figure 5, since aspect ratio simply multiplies F, the disc orientation factor, in equation (10).

[42] The rise velocities predicted by this formula are compared in Figure 5 with the velocities observed in experiments by *Gosink and Osterkamp* [1983] and *Wuebben* [1984], and with the formula (7) favored by *Morse and Richard* [2009]. The Stokes formulae sit at the lower end of data values when r is small, with for example, rise velocity in the range 2.5–3.7 mm/s depending on disc orientation predicted when radius is 1 mm. At this radius, the Stokes analysis should be marginal since Reynolds number is just greater than two, but the match with data and with Morse and Richard's high Reynolds number fit is reassuring, especially for the face-up case. At disc radii greater than 3 mm, the Stokes theory begins to deviate from data and from the Morse and Richard model.

[43] At smaller radii, the Reynolds number drops off rapidly (as the cube of r); it is Re $\sim$ 0.3, for r = 0.5 mm, and the Stokes analysis is valid for $r \leq 0.5$ mm. However, at this radius most of the experimental data plots above the two Stokes curves in Figure 5. This may be due to entrainment, where smaller discs are affected by the wake of bigger discs rising faster above them. Such entrainment effects, observed and noted in *Gosink and Osterkamp* [1983], suggest that the smaller velocities at each radius are more trustworthy, and may lead to bias in models like Morse and Richard's, which rely on a fit to all data. Such fitted models may overestimate rise velocities for smaller discs.

4.5. Turbulent Rise Velocities

[44] Fluid turbulence can alter the settling velocity of solid particles, or the rise velocity of buoyant particles and bubbles, increasing or decreasing it, depending on the relative density of the particle and on the level of turbulence [*Nielsen*, 2007]. This is because, for example, a heavy particle can spend more time in the upflow region of a vortex than in the downflow region, reducing its settling velocity. The closest analogue to frazil discs in that study is the movement of diesel drops in water, with a relative density of 0.85, which is experimental data originally from *Friedman and Katz* [2002]. Nielsen notes that smaller drops of diesel are affected by turbulence in a similar manner to particles that are denser than water, spending more time in upflow regions of vortices, enhancing diesel rise velocities by this trajectory biasing. Hence we anticipate that the diesel results are a good indication of what to expect for the case of frazil, which is slightly heavier with a relative density about 0.9.

[45] The experimental results of *Friedman and Katz* [2002] are graphed here in Figure 6. They summarize their very detailed results by noting that.

[46] 1. At (relatively) high turbulence levels all droplets approach the speed $u_*/4$, with the exception of the two lowest turbulence levels.

[47] 2. At low turbulence levels all droplets are expected to approach the still water speed.

[48] 3. At intermediate turbulence levels, the results are mixed, depending on the relative inertia of droplets (the Stokes number).

[49] The levels of turbulence in these experiments have root-mean-square fluctuation velocities that range from 30–180 mm/s, just above the range of friction velocities seen under pack ice [*McPhee et al.*, 1999]. The highest turbulence levels are the ones that approach the rise velocity $u_*/4$, illustrated by the solid straight line on the right-hand side in Figure 6. The two lower turbulence levels, likely more relevant to our situation, level out at $u_*/1.5$ and $u_*/0.7$, as illustrated by the short dashed lines in that Figure 6.

[50] After studying both Figure 6 and the figures in *Friedman and Katz* [2002] showing dependence of mean observed rise velocity $\langle V \rangle$ on droplet diameter, the rise velocity dependence on size may be summarized as:

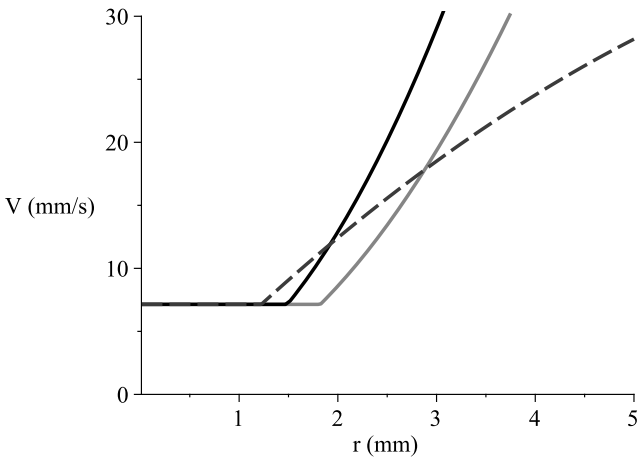

Figure 7. Modeled frazil rise velocity versus disc radius, modified by turbulence to level out at $u_*/0.7$, illustrated here for $u_* = 5$ mm/s, for Stokes theory with discs rising face-up (lower solid curve) and edge-up (upper solid curve) and for Morse and Richard's Case C (dashed line).

[51] 1. Small droplets rise faster than in still water, approaching the speed $u_*/4$ for very large u_*, but approaching $u_*/0.7$ at the lowest turbulence levels, corresponding to the very highest levels of turbulence observed under pack ice.

[52] 2. Large droplets rise at the same rate as in still water.

[53] What is interesting about smaller frazil having the same rise velocity, dependent on turbulence level rather than disc size, is that it raises the possibility of a sudden jump in the size of frazil that can settle out in a turbulent environment, that is, of a sudden collapse of mobilised frazil at a critical solid ice thickness. This sudden deposition of frazil is associated with the leveling out of the frazil rise velocity curve at a value related to turbulence levels.

[54] We consider the consequence for frazil deposition of modified rise velocities, leveling out for small frazil sizes at a minimum value of rise velocity related to u_*. The actual value is rather uncertain, but we use the value $u_*/0.7$ to illustrate the effect of turbulence (Figure 7), as it is the behavior in *Friedman and Katz* [2002] that is closest to our situation.

5. Critical Ice Thickness in Laminar Flow

[55] Here we use the rise velocities of frazil in still water; this would strictly only apply in the case of a laminar plume downflow, but turns out to be relevant to larger sized discs in a turbulent environment, and is used here to highlight the effects of turbulence in following sections. Matching the brine plume velocity with the rise velocity of frazil of radius r and aspect ratio ϵ in still water gives (in the case of Stokes flow)

$$\frac{\pi g(\varrho_w - \varrho_i)F\epsilon r^2}{8\mu} = 0.0014\left[\frac{g\varrho_{ws}\varrho_i S_w z_i\left(T_f - T_a\right)}{\varrho_w^2(2h + 0.051)}\right]^{\frac{1}{3}},$$

so that critical thickness h of fast ice, at which all frazil of radius less than or equal to r is mobilized, satisfies the equation

$$r = \sqrt{\frac{0.0112\mu}{\pi g(\varrho_w - \varrho_i)F\epsilon}}\left[\frac{g\varrho_{ws}\varrho_i S_w z_i\left(T_f - T_a\right)}{\varrho_w^2(2h + 0.051)}\right]^{\frac{1}{6}}. \qquad (11)$$

[56] This relationship is graphed in Figure 8 for discs rising face up and edge up. The graph illustrates that brine plume velocity is powerful enough to keep frazil sizes up to 1.5 mm in radii mobilized while ice thickness increases to 2 m. That is, smaller frazil sizes cannot settle out, in a nonturbulent environment, unless brine plume velocity is reduced, for example, if air temperature is much warmer than the −25°C used here for illustration.

[57] Furthermore, in these laminar flow results, the radius r below which frazil is mobilized when sea ice reaches thickness h, does not vary much with h, due to the 1/6 power in the formula. When the sea ice reaches 1 m thick, and the mixed layer is 50 m thick, then frazil with radii greater that 1.7 mm rising face-up (or 1.4 mm rising edge-up) are able to settle against the fast ice, for example, and smaller frazil remain mobilized. Changing the mixed layer thickness to 25 m or 75 m changes the critical frazil size by up to 10%, as illustrated. We also looked at a 100 m thickness, and saw a 10% increase on the 50 m results. The formula (11) shows that exactly the same changes in h result from doubling or halving the thermal drive term $T_f - T_a$, or from doubling or halving S_w. Frazil rising edge-up rises faster, and is less readily mobilized, but this only makes a 20% difference in the size of frazil that is mobilized.

[58] Since the Stokes flow assumption fails at larger frazil disc sizes, we also graph in Figure 9 the relationship between frazil radius and ice thickness that is obtained if we use Morse and Richard's rise velocity (equation (7)) instead of the Stokes values, for the case that $z_i = 50$ m and

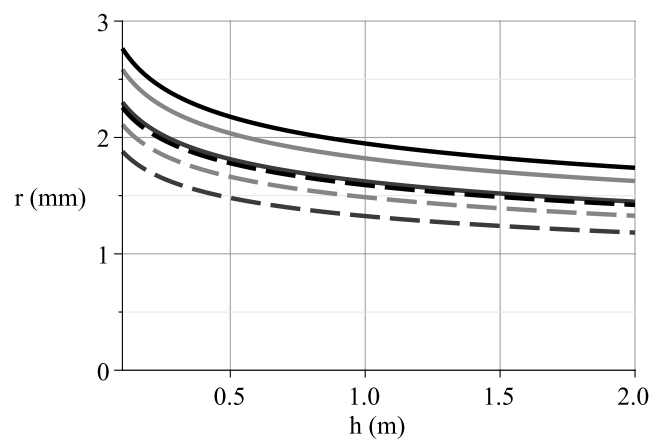

Figure 8. The maximum radius r (mm) of frazil discs that may be mobilized by brine rejection from fast ice of thickness h (m). Solid curves are for discs rising face-up, and dashed curves are for discs rising edge-up, in still water. The lower curve in each set (solid or dashed) uses a mixed layer thickness of 25 m, the middle curve 50 m, and the upper curve 75 m.

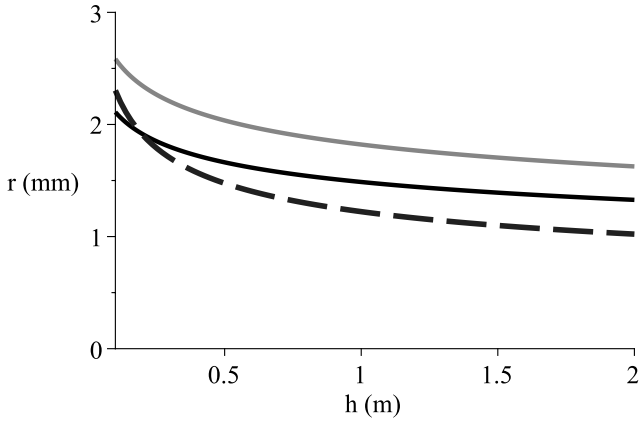

Figure 9. The maximum radius r (mm) of frazil discs that may be mobilized by brine rejection from fast ice of thickness h (m) using rise velocities in still water. A comparison of Stokes theory to *Morse and Richard*'s [2009] Case C. The upper solid curve is for Stokes flow with discs rising face-up, the lower solid curve is for Stokes flow with discs rising edge-up, and the dashed curve is for Morse and Richard's Case C.

$T_a = -25°$C. Figure 9 shows that the higher rise velocities predicted by Morse and Richard's model for smaller frazil lead to a little more frazil ice settling out, but not much more than the Stokes flow predictions. At 1 m ice thickness, the Morse and Richard model has all frazil with radii less than 1.2 mm settling out, compared to 1.4 and 1.7 mm for the two Stokes cases.

6. Critical Ice Thickness in Turbulent Water

[59] Now we illustrate the effect of a turbulence-induced cutoff in rise velocity on frazil mobilization. We use $u_* =$ 5 mm/s for illustration. Using the modified rise velocity curves illustrated in Figure 7 to match brine plume velocity leads to the frazil deposition curves in Figure 10. They are modifications of the curves in Figure 8 for the laminar case.

[60] The remarkable feature of these modified curves is the sudden deposition of most frazil sizes present, when solid ice thickness reaches the critical value corresponding to the switch of rise velocity to $u_*/0.7$. All frazil with this rise velocity is abruptly able to settle out, down to the frazil sizes mobilised by the Shields criterion, indicated by the horizontal line near the $r = 0$ axis in Figure 10.

[61] The other matter to consider when we allow for a rise velocity modified by turbulence, is whether this level of turbulence will mobilize frazil in any case, according to the Shields criterion. For example, when $u_* = 5$ mm/s, the Shields criterion says that frazil smaller than about 0.05 mm in radius will be mobilized. This should be compared to the radius at which the switch to a constant rise velocity occurs in Figure 7, approximately 1–2 mm. If u_* is larger than 15 mm/s, Figure 4 indicates that all frazil with disc radius less than 4 mm is mobilized by the Shields criterion.

[62] It is instructive to show the Shields criterion on the same graph as the frazil radius at which the catastrophic jump in rise velocity occurs, as in Figure 11. The region

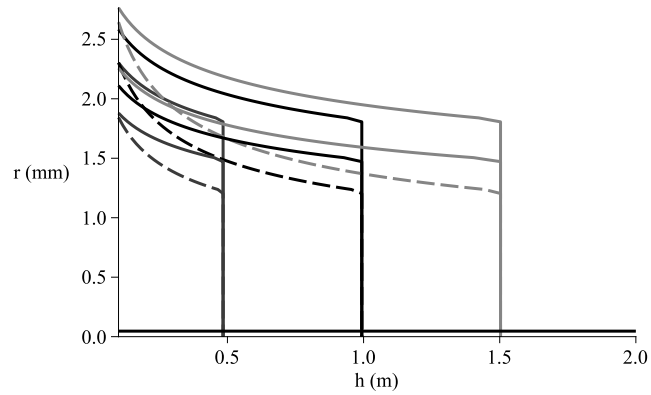

Figure 10. The maximum radius r (mm) of frazil discs that may be mobilized by brine rejection from fast ice of thickness h (m) in turbulent water. Friction velocity is 5 mm/s, and T_a is $-25°$C. The upper solid curve in each group of three is frazil rising face-up, the lower solid curve is frazil rising edge-up, and the dashed curve is Morse and Richard's Case C. The leftmost set of three curves, collapsing when turbulence dominates at $h \approx 0.5$ m, is for $z_i = 25$ m, the middle set which collapses at $h \approx 1$ m is for $z_i = 50$ m, and the rightmost set which collapses at $h \approx$ 1.5 m is for $z_i = 75$ m. The solid horizontal line at $r =$ 0.05 mm shows the Shields criterion; frazil smaller than this radius are mobilised by this level of turbulence.

between the Shields curve and the Stokes face-up curve is shaded in Figure 11. A horizontal line at a given value of u_* in the shaded region, will start at a minimum r value and end at a maximum r value, giving the range of frazil radii that are suddenly deposited when the critical ice thickness is reached where brine plume velocity matches $u_*/0.7$, using the Stokes face-up model.

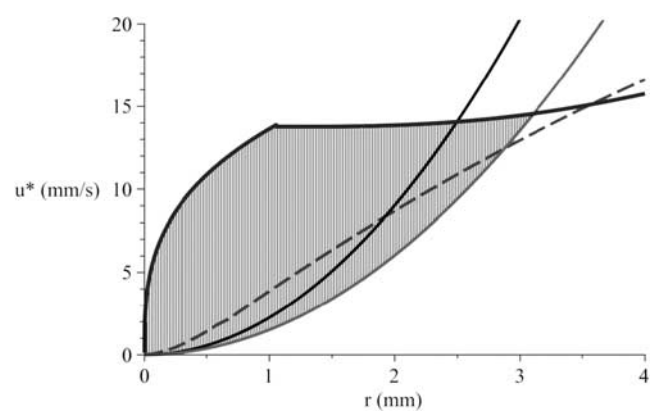

Figure 11. The Shields criterion for critical value of friction velocity (mm/s) against frazil disc radius (mm), shown on the same graph as the value of radius r where frazil rise velocity equals $u_*/0.7$. The dashed curve is for Morse and Richard's Case C, the lower solid curve is for Stokes theory for discs rising face-up, and the solid curve immediately above it is for Stokes theory for discs rising edge-up.

[63] The size of this sudden jump in frazil size deposited is largest when u_* is in the range 3–13 mm/s, values seen in the work of *McPhee et al.* [1999]. As u_* decreases, the jump size reduces and the jump is delayed to thicker sea ice and smaller frazil radii.

7. Conclusions

[64] Our results indicate that as the solid sea ice cover thickens, brine rejection slows and allows smaller and smaller frazil to settle against and freeze onto the interface. Ice cover thickness is typically of the order of 1 m for frazil of radius 1–2 mm to begin to stick to the ice cover, indicating that brine rejection is powerful enough to keep a significant size range of frazil in suspension if turbulence is ignored.

[65] Our models predict only a gradual change in the maximum size of frazil that can settle out, as ice thickness changes from 0.1 to 2 m, unless the effects of turbulence are included. Turbulence can significantly alter the frazil rise velocity curve, leading to the sudden deposition of a range of frazil sizes when conditions are right. Our calculations were for the particular case that friction velocity is 5 mm/s, but serve to illustrate the observation that frazil rise velocities are expected to be almost independent of size over a range of (smaller) frazil sizes in a turbulent environment, and that this may lead directly to a sudden transition from mobilized to settled frazil, as brine plume velocities decrease. The range of frazil sizes that experience a constant rise velocity depends on turbulence levels, and is typically all discs up to about 2 mm in radius.

[66] We have considered the effects of brine plumes on a single frazil disc. However, when a number of frazil discs accumulate, they will in turn modify the flow environment. In particular, an important mechanism that is outside the scope of this study is the stabilizing effect that accumulating buoyant frazil may have on turbulence, by altering the density gradient of the mixture of seawater and frazil ice, and by increasing its viscosity.

[67] Once enough frazil has been produced, turbulent stirring (due to both shear flow and brine rejection) near the ice-ocean interface might be stabilized sufficiently that frazil can settle against the interface, as is known to be possible with stratified cohesive sediments [*Winterwerp*, 2001], which can collapse in a positive feedback process. To study this would require a model of frazil growth and generation in the well-mixed ocean layer, coupled with a model of fast ice growth, and with a model for the vertical distribution of frazil ice in a turbulent buoyant environment. Both congelation ice growth from the fast ice layer, and frazil ice growth in suspension below, would then compete to relieve supercooling in such a model.

[68] There remains a broad question about the role and modeling of platelet ice in the growth of first-year sea ice in Antarctica. It can occupy one-third to one-half of the total landfast sea ice thickness near ice shelves. Platelet ice is clearly an important component of Antarctic sea ice, but it is not yet accounted for in sea ice growth models and remains poorly understood.

[69] **Acknowledgments.** The authors would like to thank the Marsden Fund of New Zealand, the University of Otago in New Zealand, the Korea Advanced Institute of Science and Technology in South Korea, and the Victoria University of Wellington in New Zealand for the support that enabled this work to proceed.

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